

Spring AOP Homework – Answers

1. List of Annotations: @Aspect, @Pointcut, @Before, @After, @AfterReturning, @AfterThrowing, @Around, @EnableAspectJAutoProxy, @Component, @Service, @Repository, @Controller, @RestController, @Autowired, @Bean, @Configuration.
2. Aspect Oriented Programming (AOP): AOP is a programming paradigm that separates cross-cutting concerns such as logging, security, and transactions from business logic. An aspect represents a modularized cross-cutting concern.
3. Advantages: clean code, separation of concerns, reusability. Disadvantages: proxy limitation, harder debugging.
4. Spring AOP vs Reflection vs Interceptor: Spring AOP works at method level using proxies; Reflection manipulates classes at runtime; Interceptors focus on HTTP request lifecycle.
5. Core Concepts: Aspect defines cross-cutting logic; Pointcut selects methods; JoinPoint is execution point; Advice is action taken.
6. Pointcut Declaration: Can be declared inline using execution expressions without empty methods.
7. Advice Types: Before, After, AfterReturning, AfterThrowing, Around.
8. Custom Logger: Implemented using @Around advice to log execution time, request, response, and all join points.